



P-ISSN: 2349-8528

E-ISSN: 2321-4902

www.chemijournal.com

IJCS 2025; 13(2): 01-05

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Received: 01-12-2024

Accepted: 05-01-2025

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Modern farming methods in agriculture

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Abstract

These technologies enable precise monitoring and control of environmental parameters, leading to improved crop quality and quantity in controlled environments like greenhouses. Additionally, IoT-enabled systems in hydroponics and vertical farming optimize resource consumption, enhance production efficiency, and reduce water usage compared to traditional farming methods, further contributing to the advancement of precision agriculture. Modern hydroponic systems began to take shape in the early 20th century with the work of Dr. William Frederick Gericke of the University of California, who popularized the term “hydroponics” and demonstrated its potential for large-scale crop production. Automated systems for nutrient delivery and pH control have become increasingly sophisticated, incorporating advanced algorithms and feedback loops to maintain ideal conditions for plant growth. These systems can automatically adjust nutrient concentrations and pH levels based on sensor data, reducing manual labour, and minimizing the risk of human error. Selecting high-yield and resilient crops is crucial for maximizing productivity in hydroponic systems. Crops are chosen based on their adaptability to hydroponic conditions, growth rate, and potential for high yield and quality. Leafy greens, herbs, and fruiting vegetables like tomatoes and cucumbers are commonly grown in hydroponics due to their rapid growth cycles and high market demand.

The integration of sensors and monitoring technologies has revolutionized hydroponic farming by providing real-time data on various environmental and system parameters. Sensors are employed to measure factors such as nutrient levels, pH, temperature, humidity, and light intensity, enabling precise control over the growing environment. These technologies facilitate optimal growing conditions, ensuring plant health and maximizing yield.

Keywords: Modern agriculture, data analytics, sensing technologies

Introduction

As advancements in technology introduced modern farming to alleviate the challenges of conventional soil farming, much research has been conducted and is in progress. Various farming methods, such as hydroponics, have become common practices in countries around the world. This paper aims to provide a comparative study of hydroponics and aeroponics in modern farming. Specifically, growth speed and capacity, resource efficiency, plant necessities, no environmental restrictions, and accessibility criteria are recognized for both methods (Sharma *et al.*, 2018) ^[3].

This paper aims to provide a comparative study of hydroponics and aeroponics in modern farming. Specifically, growth speed and capacity, resource efficiency, plant necessities, no environmental restrictions, and accessibility criteria are recognized for both methods (Sharma *et al.*, 2018) ^[3].

Hydroponics provides the ideal environmental settings, like factors to meet the growth of each plant element for optimum growth and performance.

There are various technologies encompassed within sensing technologies, including remote sensing techniques such as satellite imagery, aerial photography, and drones, as well as ground-based sensors like weather stations, soil moisture sensors, and nutrient sensors (Sishodia *et al.*, 2020) ^[17].

With nutrients and water in balance and roots controlling the supply of oxygen and carbon dioxide, vegetative growth of the plant is also enhanced.

Crop rotation is a sustainable farming practice that offers numerous benefits for agricultural production. It involves alternating different crops in the same area over time, which can improve soil health, increase productivity, and reduce pest and disease incidence.

Greater yields are also achieved, and downturn in crop loss is also observed with hydroponics. With nutrients and water in balance and roots controlling the supply of oxygen and carbon dioxide, vegetative growth of the plant is also enhanced.

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The fine mist covers the roots and fills the space in the propagation dome. The roots of the plants are kept about 1 inch below the sprayers, which enables the roots to dangle where they can absorb the nutrient mix. By keeping the roots air-pruned, the plant's growth is stimulated (Yang *et al.*, 2010) ^[15].

These technologies enable precise monitoring and control of environmental parameters, leading to improved crop quality and quantity in controlled environments like greenhouses. Additionally, IoT-enabled systems in hydroponics and vertical farming optimize resource consumption, enhance production efficiency, and reduce water usage compared to traditional farming methods, further contributing to the advancement of precision agriculture (Dhanaraju *et al.*, 2022) ^[11].

The quality of the products and the fast growth improve the market cycle, which is another of the most important reasons hydroponics is preferred (Dutta *et al.*, 2023) ^[10].

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They can also be used for susceptibility and hydroponic programmed times, thereby reducing the time they need to spend on these plants in oxygen and sunlight.

Aeroponics is the technology considered to be beyond hydroponics for the cultivation of numerous plants. The technique is based on sowing seedlings in a well-defined area and Suspending roots in the air; then, seedlings are regularly misted nutrients using high-frequency pressurized nozzles.

Modern farming methods use technology to increase efficiency and produce more sustainable food. These methods include

Hydroponics methods

Godfray *et al.* (2010) ^[6] Hydroponics, the practice of growing plants without soil by using mineral nutrient solutions in an aqueous solvent, represents a significant shift in agricultural methodologies. This technique, which traces its roots back to ancient civilizations like the Hanging Gardens of Babylon and the floating gardens of the Aztecs, has evolved into a sophisticated science Modern hydroponic systems began to take shape in the early 20th century with the work of Dr. William Frederick Gericke of the University of California, who popularized the term "hydroponics" and demonstrated its potential for large-scale crop production.

Barbosa *et al.* (2015) ^[21] Moreover, the rise of organic and locally-grown food movements has stimulated interest in hydroponic farming, as it can produce pesticide-free crops with a minimal environmental footprint.

Sharma *et al.* (2018) ^[3] unlike traditional soil-based agriculture methods, hydroponics offers several advantages, including the ability to precisely control nutrient levels, reduced water usage, and the elimination of soil borne diseases and pests These benefits are particularly important in the context of modern agriculture, which faces numerous challenges such as soil degradation, reduction in cultivated area.

Lakhia *et al.* (2018) ^[5] Innovations include the use of modular systems that allow simple expansion and the integration of sensors to monitor nutrient levels, pH, and temperature, facilitating precise control over the growing environment.

Technological advancements in hydroponic

Automation and Control Systems

Shi *et al.* (2018) ^[4] Advancements in sensor technology have led to the development of wireless and low-power sensors, which enhance the efficiency and scalability of hydroponic systems.

Resh (2013) ^[2] the integration of sensors and monitoring technologies has revolutionized hydroponic farming by providing real-time data on various environmental and system parameters. Sensors are employed to measure factors such as nutrient levels, pH, temperature, humidity, and light intensity, enabling precise control over the growing environment. These technologies facilitate optimal growing conditions, ensuring plant health and maximizing yield.

Automated Nutrient Delivery and pH Control

Grisso *et al.* (2011) ^[1] Additionally, advancements in precision agriculture have led to the development of variable rate technology, which allows for the customized application of nutrients to different zones within a hydroponic system, further optimizing resource use and crop performance.

Domingues *et al.* (2012) ^[22] Automated systems for nutrient delivery and pH control have become increasingly sophisticated, incorporating advanced algorithms and feedback loops to maintain ideal conditions for plant growth. These systems can automatically adjust nutrient concentrations and pH levels based on sensor data, reducing manual labour, and minimizing the risk of human error.

LED Lighting and Photoperiod Management

Bourget (2008) ^[23] the development of light-emitting diode (LED) technology has transformed the lighting strategies used in hydroponics. LEDs offer several advantages over traditional lighting sources, including energy efficiency, long lifespan, and the ability to emit specific wavelengths of light that are optimal for plant growth.

Mitchell *et al.* (2015) ^[24] Modern LED systems can be programmed to deliver precise light cycles, adjusting intensity and spectral output to match the needs of different crops and growth phases. This flexibility enhances photosynthesis and promotes healthier, faster-growing plants.

IoT and Smart Farming

Hu *et al.* (2021) ^[25] The Internet of Things (IoT) has significantly impacted hydroponic farming by enabling the seamless integration of various devices and systems. IoT devices, such as smart sensors, automated controllers, and connected cameras, provide a comprehensive overview of the hydroponic environment, allowing for remote monitoring and control.

Liang *et al.* (2023) ^[26] the use of IoT in hydroponics extends to data collection and analysis, where connected devices continuously gather information that can be used to improve system performance and predict potential issues

Data Analytics and Machine Learning

Kamilaris *et al.* (2018) ^[27] ML algorithms can predict crop growth rates, nutrient requirements, and potential disease outbreaks, allowing for proactive interventions.

Tantalaki, *et al.* (2019) ^[28] Data analytics and machine learning (ML) applications have emerged as powerful tools in hydroponic farming, offering insights that drive decision-making and system optimization. By analysing data collected from IoT devices, growers can identify patterns and correlations that inform better management practices.

Disease Management and Pest Control

Harman *et al.* (2004) ^[29]: Effective disease management is crucial for maintaining plant health and ensuring high yields in hydroponic systems. Innovations in disease prevention and treatment include the use of biological control agents, such as beneficial bacteria and fungi, which can suppress pathogenic microorganisms and promote plant health.

Integrated Pest Management Strategies

Bale *et al.* (2008) ^[30] integrated Pest Management (IPM) strategies in hydroponic systems combine biological, cultural, mechanical, and chemical methods to control pest populations and reduce reliance on synthetic pesticides and Biological control, using natural predators and parasitoids, is a cornerstone of IPM, providing effective and environmentally friendly pest management solutions.

Selection of High-Yield and Resilient

Sharma *et al.* (2018) ^[3] Selecting high-yield and resilient

crops is crucial for maximizing productivity in hydroponic systems. Crops are chosen based on their adaptability to hydroponic conditions, growth rate, and potential for high yield and quality. Leafy greens, herbs, and fruiting vegetables like tomatoes and cucumbers are commonly grown in hydroponics due to their rapid growth cycles and high market demand.

Innovations in Vertical Hydroponic

Kalantari *et al.* (2017) ^[31] recent innovations in vertical hydroponic systems focus on integrating automation and smart technologies to enhance productivity and sustainability. Automated systems for nutrient delivery, lighting, and environmental control ensure optimal growing conditions at all levels of the vertical structure.

Kalantari *et al.* (2018) ^[32] Lightweight, durable materials and innovative design solutions are making it easier to implement vertical farming in a variety of settings, from urban rooftops to indoor facilities.

Difference between hydroponic and aeroponic agriculture

Aspect	Hydroponic	Aeroponic
Growing Medium	Utilizes inert materials (like perlite, Rockwool, or coconut coir) as support for plant roots within a nutrient-rich water solution.	Does not use a growing medium; plant roots are suspended in air and misted with a nutrient rich solution.
Method of Nutrient Delivery	Nutrients delivered through the water solution that constantly circulates around the roots.	Nutrients are delivered directly to the roots via a fine mist or spray.
Water Usage	Generally uses more water compared to aeroponics due to the continuous flow or recirculation of water.	Highly water-efficient, using significantly less water compared to hydroponics due to the misting system and recycling capabilities.
Oxygen Levels for Roots	Good oxygen levels but not as high as aeroponics because roots are submerged in water.	Provides the highest oxygen levels to the roots as they are exposed to air, leading to faster growth rates.
Plant Growth	Growth rates are typically slower than aeroponics due to the indirect nutrient delivery system.	Offers faster growth rates and increased yields owing to direct nutrient delivery and optimal oxygenation of roots.
Susceptibility to Disease	More prone to diseases spread through the water system or growing medium.	Less prone to diseases due to the absence of a growing medium and exposure to open air.
Complexity of System	Relatively simpler system compared to aeroponics, easier for beginners to set up and maintain.	More complex system requiring precise misting, potentially challenging for novices without technical expertise.

(Source: Shashank *et al.*, 2024) ^[33]

Aquaponics

Rakocy *et al.* (2016) ^[34] Aquaponic system is based on the principle that it utilizes the effluent water that is produced by the fish which is a valuable source of nutrients for plants.

Wongkiew *et al.* (2017) ^[35] the fish tank, biofilter, and hydroponic system are all connected via a water supply.

Palm *et al.* (2018) ^[36] these definitions have evolved significantly over time, reflecting advancements in both scientific understanding and practical applications. Early definitions by Rakocy (1994) ^[16] focused on integrating hydroponics with fish culture to increase production efficiency, with limited discussion on commercial viability.

Greenfeld *et al.* (2019) ^[37] the concept of aquaponics has gained popularity around the world as a sustainable method of food production. Greenfeld *et al.* (2021) ^[37] it is maintained with the minimum number of chemical additives and is antibiotic-free, thus supporting the nutrient cycle of the bacteria. Gallo *et al.* (2022) ^[39] Aquaponic systems have much higher water use efficiency than conventional systems.

Vasdravanidis *et al.* (2022) ^[40] It has been demonstrated that innovations such as aquaponics can effectively mitigate climate change's impact on aquaculture, especially freshwater fish like rainbow trout (*Oncorhynchus mykiss*), which are co-cultivated with plants in a sustainable manner under controlled abiotic conditions.

Benefit of Aquaponics

Joyce *et al.* (2019) ^[41] by enabling farming closer to urban areas, aquaponics can reduce transportation costs and the carbon footprint of food production.

Lobillo-Eguibar *et al.* (2020) ^[42] Aquaponic systems serve various purposes, including commercial ventures, educational initiatives, hobbies, research endeavors, or providing self-sustaining food production for families Small-scale aquaponic systems (SAS) are designed to provide affordable, sustainable solutions for individual households or small communities.

Babatunde *et al.* (2023) ^[43] SAS can offer valuable social benefits, such as promoting food security, reducing food miles, and fostering community engagement in sustainable practices found that the economic viability of small-scale systems depends on optimizing the fish-to-plant revenue ratio, with studies suggesting a minimum of 30:70 for profitability. Chung-Do *et al.* (2024) ^[44] conducted a detailed benefit analysis of an aquaponic system and drew out some important social and economic benefits. Increasing food security and food sovereignty is a significant social benefit of aquaponic systems Employment and income generation are some of the main economic benefits.

Vertical farming

Shamshiri *et al.* (2018) ^[45] Vertical farming is a modern approach

to agriculture where crops are grown indoors, stacked vertically. This method uses hydroponics or aeroponics (growing plants without soil) and offers several advantages. It maximizes space, uses less water, and reduces the need for pesticides and herbicides.

Eldridge *et al.* (2020) ^[46] The concept of vertical farming was first proposed in the 1980s by a Swedish ecological farmer, with the idea of growing vegetables in urban areas (Vertical farms can be set up almost anywhere, even underground, enabling local food production and delivering fresh, nutritious food throughout the year).

Kalantari *et al.* (2021) ^[47] Vertical farming has emerged as a transformative approach in modern agriculture, particularly in addressing the challenges of urban food production, sustainability, and resource efficiency.

The concept of vertical farming is not entirely new, but its practical implementation has gained momentum in recent years due to advances in technology and a greater emphasis on sustainable food production.

Role of vertical hydroponics in urban farming

Searchinger *et al.* (2018) ^[48] Vertical farming presents a promising solution to this challenge. By leveraging controlled indoor environments, vertical farming mitigates societal environmental and economic threats to food security.

Khan *et al.* (2020) ^[49] Optimized conditions for temperature, humidity, CO₂, and lighting can significantly shorten growing cycles, increase plant density, and ultimately enhance.

Mir *et al.* (2022) ^[50] the benefits of vertical farming extend beyond increased production. This technology fosters eco-friendly and sustainable food production by reducing energy and water consumption, minimizing pollution, and improving access to fresh, nutritious food.

Soilless Cultivation Techniques

Arcas-Pilz *et al.* (2021) ^[51] Soilless farming methodologies form the bedrock of vertical agriculture in urban areas. These methods encompass diverse systems like Nutrient Film Technique (NFT), aeroponics, and drip irrigation, each offering distinct advantages suited to specific plant varieties and spatial limitations.

Samal *et al.* (2024) ^[52] this sustainable and synergistic system integrates fish farming (aquaculture) with hydroponics. Nutrient-rich fish waste provides sustenance for the plants, while the plants filter the water for the fish, creating a closed-loop ecosystem. Minimal monitoring is required after initial setup.

Artificial Lighting Systems

Wong *et al.* (2020) ^[53] Due to limited access to natural sunlight, indoor vertical farms rely heavily on artificial lighting for plant growth. Light-Emitting Diodes (LEDs) are the primary source of illumination in these systems. Ongoing research focuses on optimizing LED light characteristics, including intensity, spectrum, and photoperiod, for various crops.

Automation and Monitoring Systems

Kacira *et al.* (2020) ^[54] Incorporating automation for tasks such as nutrient delivery, lighting control, and environmental monitoring can significantly enhance the efficiency and manageability of vertical hydroponic systems. Sensor technology and the Internet of Things (IoT) enable real-time monitoring and adjustments, optimizing resource utilization and plant growth.

Technological Advancements

Valladares *et al.* (2019) ^[55]: Continued research and development in hydroponic technologies are expected to improve efficiency, reduce costs, and make systems easier to use. Advancements in automation, lighting, and nutrient delivery will be crucial.

Applications of Drones in Agriculture

The application of drone in almost all areas of agriculture, making it more productive and efficient while reducing waste to its resources.

1. **Crop Monitoring:** Drone captures high-resolution images and multispectral data to facilitate the detection of any stress, nutrient deficiency, and pests arising in the crop, thus allowing for timely intervention.
2. **Precision Agriculture:** Drones can help prepare detailed aerial maps that farmers can use to optimize resource allocation based on variability in soils, therefore increasing yields and minimizing environmental impacts.
3. **Irrigation Management:** Thermal sensors enabled drones can sense an infringement of a temperature difference as an indication of irrigation, and assist in effective usage and saving.
4. **Crop Spraying:** Drones enable fast uniform coverage from a view point of aerial spray for pesticides and fertilizers, where they can reach remote areas and limit contact with chemicals to handlers. Optimization of the parameters of various measurement methods helps enhance the UAV-based spraying system.
5. **Soil Health Monitoring:** Drones produce NDVI maps and determine soil moisture and nutrients that drive the upgrading of soil management and crop yields.

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